MILITARY SPECIFICATION

AMPLIFIER, HYBRID, RADIO FREQUENCY FOR XM122 FIRING DEVICE RECEIVER

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of hybrid radio frequency amplifier for the XM122 firing device receiver.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of inactivation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test For
MIL-STD-810 - Environmental Test Methods
MIL-STD-883 - Test Methods and Procedures For Microelectronics
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9278755 - U1 R.F. Amplifier, Hybrid

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement fucntions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies.-The components and asembly shall comply with all requirements specified in drawing 9278755 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Vibration.-The module shall not show evidence of damage or failure when tested in accordance with 4.5.1.

3.4 Rough Handling.-The module shall not show evidence of damage or failure when tested in accordance with 4.5.2.

3.5 Humidity.-The module shall not show evidence of damage or failure when tested as specified in 4.5.3.

3.6 Temperature Cycling.-The module shall withstand sudden changes in temperature when tested as specified in 4.5.4.

3.7 Gross leak.-The module shall not shown evidence of leakage when tested as specified in 4.5.5.

3.8 Fine leak.-The module shall not show evidence of leakage when tested as specified in 4.5.6.

3.9 Electrical Performance Characteristics

3.9.1 Voltage gain.-The module voltage gain shall be eighteen and five tenths plus or minus one decibel (18.5 db $\pm$ 1 db) when tested as specified in 4.5.7.1

3.9.2 Noise figure.-The overall noise figure shall be eight decibel (8 db) maximum when tested as specified in 4.5.7.2.

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3.9.3 Input impedance.-The input impedance between pin numbers two and three (2 and 3) shall be thirty to fifty (30 to 50) ohms when tested as specified in 4.5.7.3.

3.9.4 Output impedance.-The output impedance between pins six and two (6 and 2) shall be twenty two picifarads (22 pfd) plus or minus five percent (± 5%) when tested as specified in 4.5.7.4.

3.9.5 Bias voltage.-The module shall generate a voltage of one and forty hundreds plus or minus five hundreds volts direct current (1.40 ± 0.05 VDC) when tested as specified in 4.5.7.5.

3.9.6 Current drain.-The module current drain shall not exceed two milliamperes (2 ma) when tested as specified in 4.5.7.6.

3.9.7 Voltage variations.-The module shall meet the requirements of 3.9.1 through 3.9.6 when all referenced voltages are varied to the specified maximum and minimum limits as tested in 4.5.7.7.

3.9.8 Temperature variations.-The module shall meet the requirements of 3.9.7 over the temperature range of minus thirty two degrees centigrade (-32°C) to plus sixty five degrees centigrade (+65°F) when tested as specified in 4.5.7.8.

3.10 Rework or retest.-Any module failing to comply with all of the test requirements and undergoes rework or repair shall be tested as specified in 4.5.10.

3.11 Test temperature.-Unless specified otherwise, all testing shall be performed at a temperature of twenty degrees centigrade plus ten degrees centigrade (20°C ± 10°C).

3.12 Data requirements.-Data shall be recorded and retained for all test required by this specification and as specified in 4.5.9.

3.13 Workmanship.-The module shall be free from dirt, chips, grease, flux, rust or other foreign matter that may impair proper functioning. The cleaning methods used shall not be injurious to any part or assembly nor shall the parts or assembly be contaminated.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspections.

4.3 First article inspection

4.3.1 Submission.-The contract shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of twenty five (25) modules.

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63117 (AR)

PARAGRAPH	TITLE Amplifier, Hybrid, Radio Frequency For XM122 Firing Device Receiver			SHEET 1 OF 1		DRAWING NUMBER 9278755 / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST		NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects		25		3.2	4.4.2.1
	Testing	(a)				
	Vibration	(a)	25		3.3	4.5.1
	Rough Handling	(a)	25		3.4	4.5.2
	Humidity	(a)	25		3.5	4.5.3
	Temperature Cycling	(a)	25		3.6	4.5.4
	Gross Leak	(a)	25		3.7	4.5.5
	Five leak	(a)	25		3.8	4.5.6
	Gain	(a)	25		3.9.1	4.5.7.1
	Noise figure	(a)	25		3.9.2	4.5.7.2
	Input impedance	(a)	25		3.9.3	4.5.7.3
	Output impedance	(a)	25		3.9.4	4.5.7.4
	Bias voltage	(a)	25		3.9.5	4.5.7.5
	Current drain	(a)	25		3.9.6	4.5.7.6
	Voltage variations	(a)	25		3.9.7.3	4.5.7.7
	Temperature variations	(a)	25		3.10	4.5.7.8

NOTES: (a) Testing shall be performed as indicated in sequence shown in Table I.

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4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. - See MIL-A-47078.

a. Sampling plans. - Unless otherwise specified in the classification of defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1	Amplifier, Hybrid, Radio Frequency For XM122 Firing Device Receiver		1 of 1	927855
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST		NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined					
Major						
101.	Pin location			0.40%	3.2	Gage
102.	Minimum pin length			0.40%	3.2	Gage
103.	Assembly damaged or distorted			0.40%	3.2	Visual
	Mechanical Tests					
104.	Vibration	(a)		100%	3.3	4.5.1
105.	Rough Handling	(a)		100%	3.4	4.5.2
106.	Humidity	(a)		100%	3.5	4.5.3
107.	Temperature cycling	(a)		100%	3.6	4.5.4
108.	Gross leak	(a)		100%	3.7	4.5.5
109.	Five leak	(a)		100%	3.8	4.5.6

NOTES:
Testing shall be in the sequence shown.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.1					9278775
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electrical Performance Tests				
110.	Gain (a)		100%	3.9.1	4.5.7.1
111.	Noise figure (a)		100%	3.9.2	4.5.7.2
112.	Input impedance (a)		100%	3.9.3	4.5.7.3
113.	Output impedance (a)		100%	3.9.4	4.5.7.4
114.	Bias voltage (a)		100%	3.9.5	4.5.7.5
115.	Current drain (a)		100%	3.9.6	4.5.7.6
116.	Voltage variation (a)		100%	3.9.7	4.5.7.7
117.	Temperature variation (a)		100%	3.10	4.5.7.8
118.	Data requirements (a)		100%	3.12	4.5.9
	Minor:				
201.	Pin diameter		0.65%	3.2	Gage
202.	Outer flange diameter		0.65%	3.2	Gage
203.	Evidence of poor workmanship		1.0%	3.13	Visual

NOTES:

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4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations, and tests prescribed herein is described in the Paragraph Reference Inspection Method column in the tables of paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Vibration.-The module shall be subjected to the vibration test requirements of Method 514, Procedure XI, Part I, vibration and Part 2, Bounce, Loose Cargo of MIL-STD-810, (Non-destructive test).

4.5.2 Rough handling.-The module shall be subjected to the requirements of paragraph 5.3, Test No. 114 of MIL-STD-331 (Non-destructive test).

4.5.3 Humidity test.-The module shall be subjected to Procedure I, Method 507 of MIL-STD-810 (Non-destructive test).

4.5.4 Temperature cycling.-The module shall be subjected to ten (10) temperature cycles, in accordance with Condition C, Method 1010 of MIL-STD-883 (Non-destructive test).

4.5.5 Gross leak.-The module shall be subjected to the gross leakage requirements of Condition C, Method 1014 of MIL-STD-883. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.6 Fine leak.-The module shall be subjected to the fine leakage requirements of Condition A, Method 1014 of MIL-STD-883. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.7 Electrical Performance Characteristics

4.5.7.1 Voltage gain.-The module shall be measured for the specified voltage gain using Test Circuit No .1. Test frequency specified voltage gain using Test Circuit No. 1. Test frequency minus one thousand hertz (25,500,000 Hz ± 1000 Hz). The input shall be driven from a fifty (50) ohm source at minus forty five decibel (-45 dbm) maximum. Transformer, T1, shall be adjusted for maximum output. The RF output of the test circuit shall be terminated into fifty (50) ohms. Pin number four (4) shall have

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an applied bias voltage of one and one hundred and fifty five thousands plus or minus five thousands volts direct current (1.155 VDC ± 0.005 VDC). Pin number eight (8) shall have an applied supply voltage of three and twenty five hundreds plus or minus twenty five hundreds (3.25 ± 0.25) VDC. Pin number two (2) shall be grounded. Any module failing to comply with all of the specified requirements shall be classed defective. (Non-destructive test).

4.5.7.2 Noise figure.-The module shall be tested for the specified noise level using Test Circuit No. 1. Test frequency shall be twenty five million five hundred thousand hertz plus or minus ten thousand hertz (25,500,000 Hz ± 10,000 Hz). Pin number four (4) shall have an applied bias voltage of one and one hundred and fifty five thousands plus or minus five thousands (1.155 VDC ± 0.005 VDC). Pin number eight (8) shall have an applied supply voltage of three and twenty five hundreds plus or minus twenty five hundreds (3.25 VDC ± 0.25 VDC). Pin number two (2) shall be grounded. A noise figure meter having a fifty (50) ohm source impedance shall be connected to the RF input. A fifty (50) ohm load impedance shall be connected to the output of the test circuit. Any module failing to comply with all of the specified requirements shall be classed defective. (Non-destructive test).

4.5.7.3 Input impedance.-The module shall be measured for the specified input impedance requirements using Test Cirucit No. 1. Pin two (2) of the module shall be internally connected to the case. Test frequency shall be twenty five million, five hundred thousand hertz plus or minus ten thousand hertz (25,500,000 Hz ± 10,000 Hz). Pin number four (4) shall have an applied bias voltage of one and one hundred and fifty five thousands plus or minus five thousands (1.155 VDC ± 0.005 VDC). Pin number eight (8) shall have an applied supply voltage of three and twenty five hundreds plus or minus twenty five hundreds (3.25 VDC ± 0.25 VDC). A vector impedance meter shall be connected at the RF input to read the module characteristics. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.7.4 Output impedance.-The module shall be measured for the specified requirements using Test Circuit No. 2. Test frequency shall be twenty five million five hundred thousand hertz plus or minus twenty thousand hertz (25,500,000 Hz ± 50,000 Hz).

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Pin number four (4) shall have an applied bias voltage of one and one hundred and fifty five thousands plus or minus five thousands. (1.155 VDC ± 0.005 VDC). Pin number eight (8) shall have an applied supply voltage of three and twenty five hundreds plus or minus twnety five hundreds (3.25 VDC ± 0.25 VDC). A reactance meter shall be used to read the module characteristics. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.7.5 Bias voltage.-The module shall meet the specified requirements when using Test Circuit No. 3. Measurements shall be performed with a twelve thousand plus or minus five hundred (12,000 ± 500) ohm load between pin five (5) and ground. Pin one (1) shall have an applied voltage of nine to thirteen and five tenths (9.0 VDC to 13.5 VDC). Pin number eight (8) shall have an applied voltage of three and twenty five hundreds plus or minus twenty five hundreds (3.25 VDC ± 0.25 VDC). Pin five (5) shall not generate an output when pin one (1) is held at zero (0) VDC. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.7.6 Current drain.-The module shall be tested for the specified requirements using Test Circuit No. 4. Measurements shall be made on pin number eight (8) using approved equipment. Pin number four (4) shall have an applied bias input voltage of one and one hundred and fifty five plus or minus five thousands (1.155 VDC ± 0.005 VDC). Pin number eight (8) shall have an applied voltage of three and twenty five hundreds plus or minus twenty five hundreds (3.25 VDC ± 0.25 VDC). A twelve thousand plus or minus five hundred (12,000 ± 500) ohm resistive load shall be connected between pins five and two (5 and 2). Pin number one (1) shall have an applied voltage of nine to thirteen and five tenths (9.0 to 13.5) VDC. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.7.7 Voltage variations.-The module shall be tested for the specified requirements using Test Circuits 1 through 4 as applicable for the respective test. Voltages shall be varied to the limits in previous test of 4.5.7.1-4.5.2.6. Any module failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

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4.5.7.8 Temperature variations.-The module shall be tested for the specified requirements over the specified temperature range using the respective test circuits. The module's internally generated bias voltage shall have a negative temperature coefficient of four and nine tenths to six and seven tenths millivolts per degree centigrade (4.9 mv/°C to 6.7 mv/°C) when pin number five (5) is loaded with a twelve thousand plus or minus five hundred (12,000 ± 500) ohm resistive load. Readings shall be taken at temperatures of minus thirty two degrees centigrade (-32°C) minimum, zero degrees centigrade plus or minus five degrees centigrade (0°C ± 5°C) plus twenty five degrees centigrade plus or minus five degrees centigrade (+ 25°C ± 5°C) and a minimum temperature of plus sixty five degrees centigrade (+65°C). Voltage variations shall be as stated except for the biasing voltage whose limits are established herein. Any module failing to comply with all of the specified requirements shall be classed defective (Non-defective test).

4.5.8 Rework or retest.-Any module failing to meet all of the requirements and is reworked or repaired shall be subject to all of the test performances of 4.5.1 through 4.5.7.8 before being classed acceptable. Any module failing to comply with all of the requirements after rework shall be classed defective. All rework or repair shall be documented as to cause of rejection and the work performed. The rejected data shall accompany the data for acceptance of the module (Non-destrutive test).

4.5.9 Data requirements.-The contractor shall submit a quality inspection test, demonstrated and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving all detailed inspection and test results.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Each module shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C.-The module, preserved and packaged in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking.-Marking of the carton shall comply with MIL-STD-1169.

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5.4 Shipping. - When modules of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The assemblies covered by this specification are intended for use as an RF amplifier for the M122 firing device receiver. This specification is to be used only as a guideline in establishing on updated version of this device.

6.2 Ordering data. - See Section 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval (See MIL-A-48078). - Submit equipment designs as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

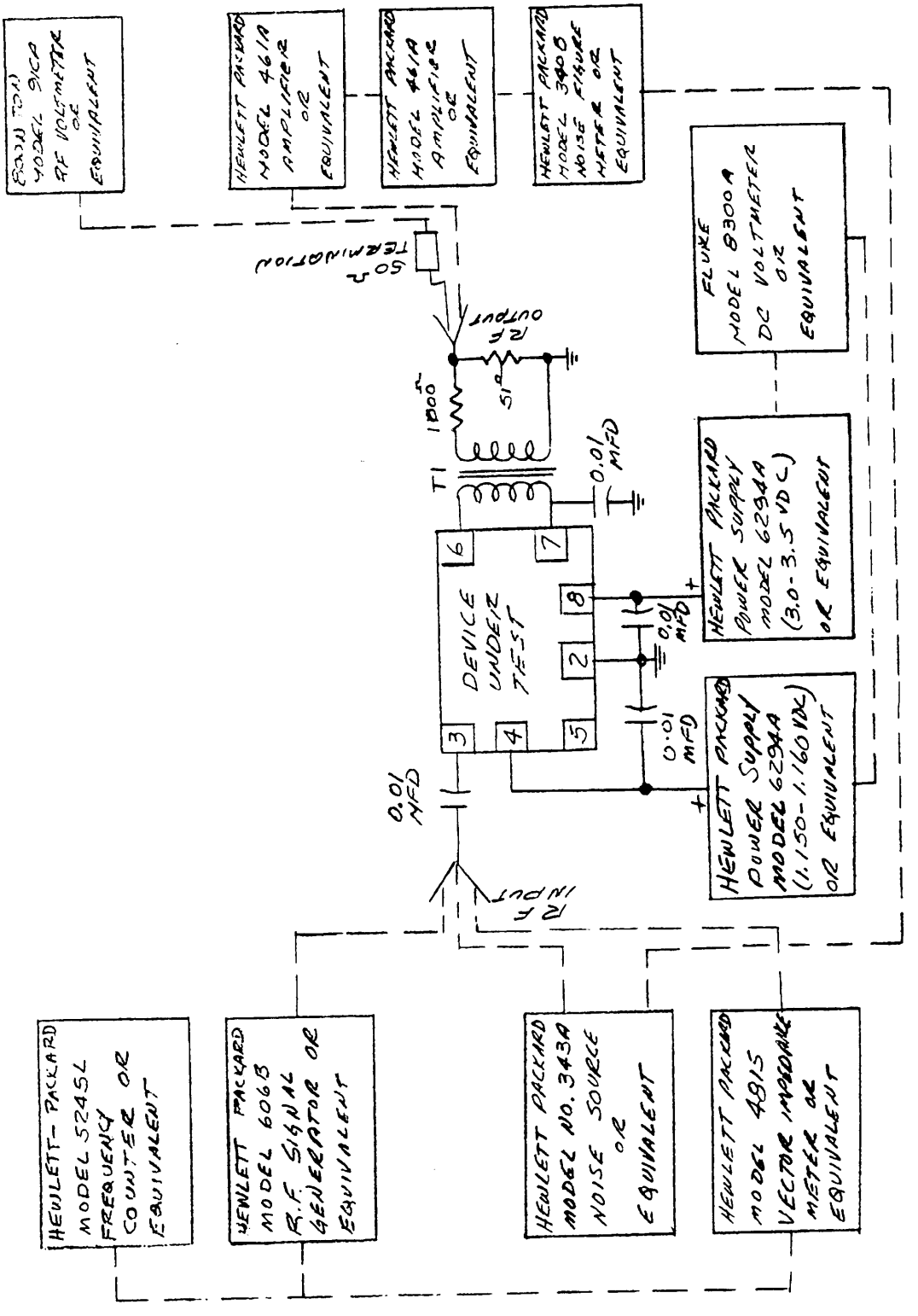
6.4 Distribution of ammunition data cards, inspection results and test data. - Distribution of ammunition data cards, inspection and test results shall be submitted to : Commander, U.S. Army Armament Research and Development Command,ATTN: DRDAR-QAA-Q, Dover, New Jersey 07801.

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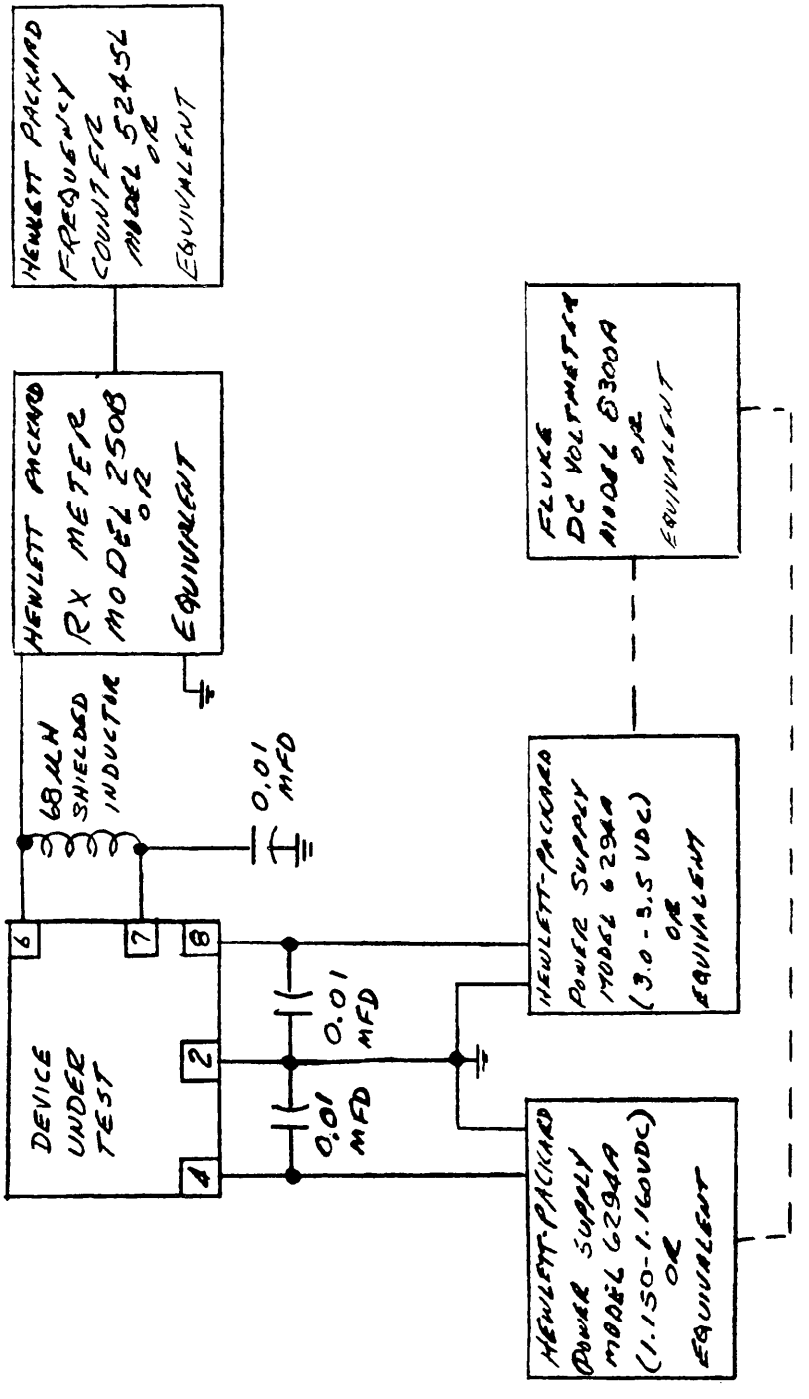
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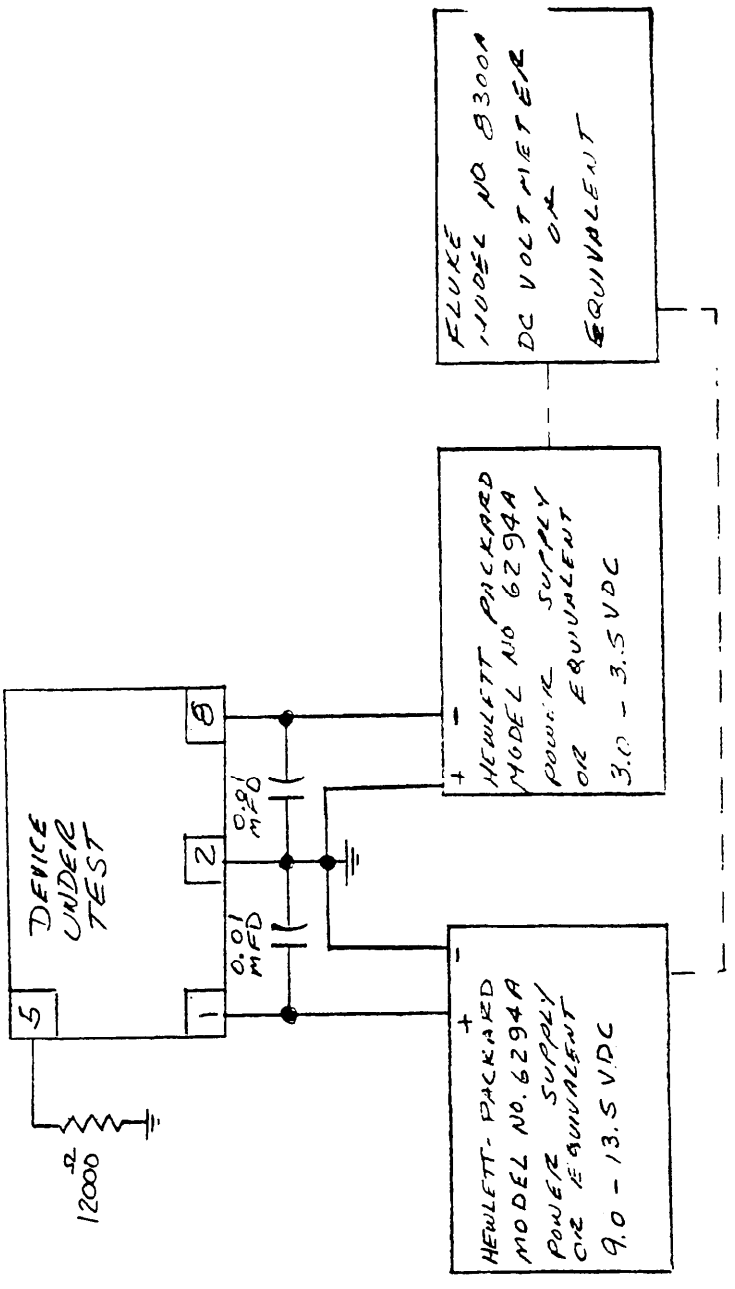
TEST CIRCUIT NO.1

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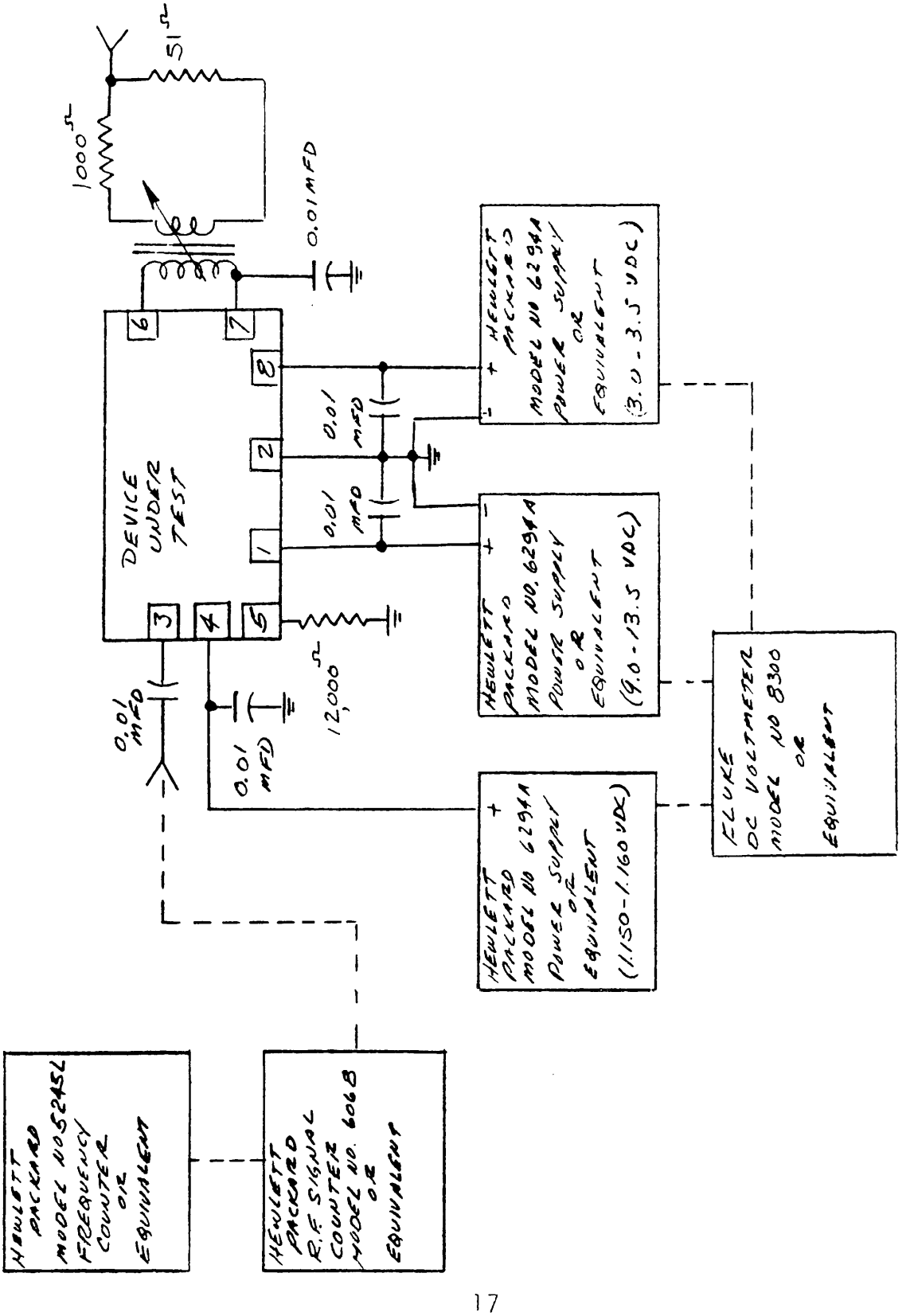
TEST CIRCUIT NO. 2

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TEST CIRCUIT NO. 3

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TEST CIRCUIT NO. 4

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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